BBM 202 - ALGORITHMS

HACETTEPE UNIVERSITY

DEPT. OF COMPUTER ENGINEERING

DATA COMPRESSION

May. 15, 2017

Acknowledgement: The course slides are adapted from the slides prepared by R. Sedgewick and K. Wayne of Princeton University.

Data Compression

- Run-length coding
- Huffman compression

Data compression

Compression reduces the size of a file:

- To save space when storing it.
- To save time when transmitting it.
- Most files have lots of redundancy.

Who needs compression?

- Moore's law: # transistors on a chip doubles every 18-24 months.
- Parkinson's law: data expands to fill space available.
- Text, images, sound, video, …

> *" Everyday, we create 2.5 quintillion bytes of data—so much that 90% of the data in the world today has been created in the last two years alone. "* — *IBM report on big data (2011)*

Basic concepts ancient (1950s), best technology recently developed.

Applications

Generic file compression.

- Files: GZIP, BZIP, 7z.
- Archivers: PKZIP.
- File systems: NTFS, HFS+, ZFS.

Multimedia.

- Images: GIF, JPEG.
- Sound: MP3.
- Video: MPEG, DivX™, HDTV.

Communication.

- ITU-T T4 Group 3 Fax.
- V.42bis modem.
- Skype.

Databases. Google, Facebook, ...

Google™ facebook

Lossless compression and expansion

uses fewer bits (you hope)

Message. Binary data B we want to compress.

Compress. Generates a "compressed" representation $C(B)$.

Expand. Reconstructs original bitstream B.

Basic model for data compression

Compression ratio. Bits in $C(B)$ / bits in B.

Ex. 50-75% or better compression ratio for natural language.

Food for thought

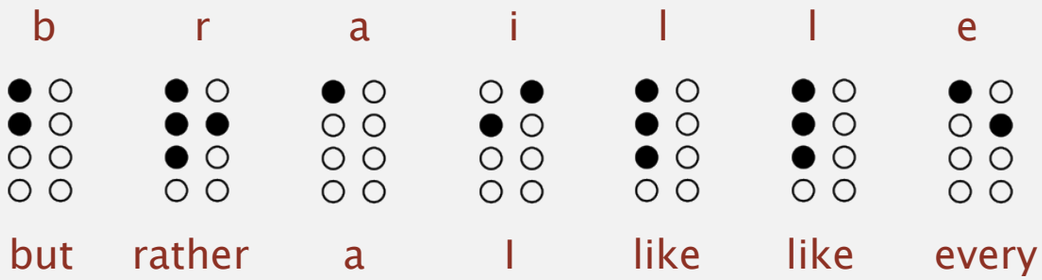
Data compression has been omnipresent since antiquity:

- Number systems.
- Natural languages.
- Mathematical notation.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

has played a central role in communications technology,

- Grade 2 Braille.
- Morse code.
- Telephone system.



and is part of modern life.

- MP3.
- MPEG.

Q. What role will it play in the future?

Data representation: genomic code

Genome. String over the alphabet { A, C, T, G }.

Goal. Encode an N-character genome: `ATAGATGCATAG`...

Standard ASCII encoding.

- 8 bits per char.
- $8\ N$ bits.

char	hex	binary
A	41	01000001
C	43	01000011
T	54	01010100
G	47	01000111

Two-bit encoding.

- 2 bits per char.
- 2 N bits.

char	binary
A	00
C	01
T	10
G	11

Fixed-length code. k-bit code supports alphabet of size 2^k.

Amazing but true. Initial genomic databases in 1990s used ASCII.

Reading and writing binary data

Binary standard input and standard output. Libraries to read and write bits from standard input and to standard output.

public class BinaryStdIn	
boolean readBoolean()	*read 1 bit of data and return as a* `boolean` *value*
char readChar()	*read 8 bits of data and return as a* `char` *value*
char readChar(int r)	*read* `r` *bits of data and return as a* `char` *value*
[*similar methods for* `byte` *(8 bits);* `short` *(16 bits);* `int` *(32 bits);* `long` *and* `double` *(64 bits)*]	
boolean isEmpty()	*is the bitstream empty?*
void close()	*close the bitstream*

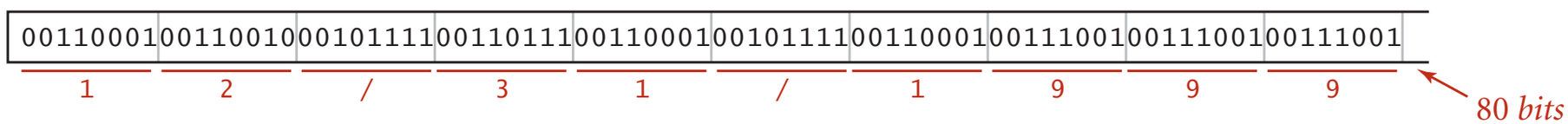
public class BinaryStdOut	
void write(boolean b)	*write the specified bit*
void write(char c)	*write the specified 8-bit* `char`
void write(char c, int r)	*write the* `r` *least significant bits of the specified* `char`
[*similar methods for* `byte` *(8 bits);* `short` *(16 bits);* `int` *(32 bits);* `long` *and* `double` *(64 bits)*]	
void close()	*close the bitstream*

Writing binary data

Date representation. Three different ways to represent 12/31/1999.

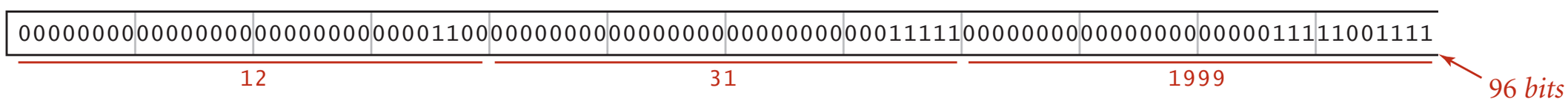
A character stream (StdOut)

```
StdOut.print(month + "/" + day + "/" + year);
```



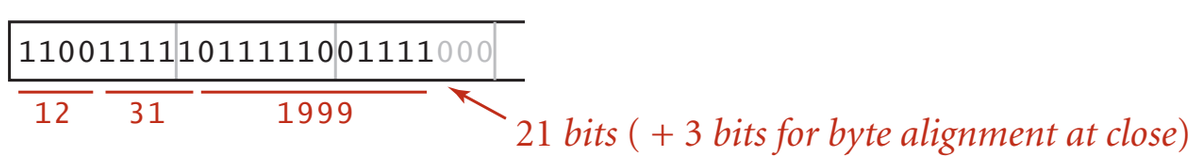
Three ints (BinaryStdOut)

```
BinaryStdOut.write(month);
BinaryStdOut.write(day);
BinaryStdOut.write(year);
```



A 4-bit field, a 5-bit field, and a 12-bit field (BinaryStdOut)

```
BinaryStdOut.write(month, 4);
BinaryStdOut.write(day, 5);
BinaryStdOut.write(year, 12);
```



Binary dumps

Q. How to examine the contents of a bitstream?

Standard character stream

```
% more abra.txt
ABRACADABRA!
```

Bitstream represented as 0 and 1 characters

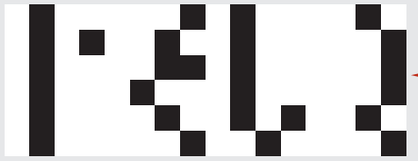
```
% java BinaryDump 16 < abra.txt
0100000101000010
0101001001000001
0100001101000001
0100010001000001
0100001001010010
0100000100100001
96 bits
```

Bitstream represented with hex digits

```
%  java HexDump 4 < abra.txt
41 42 52 41
43 41 44 41
42 52 41 21
12 bytes
```

Bitstream represented as pixels in a `Picture`

```
% java PictureDump 16 6 < abra.txt
```



16-by-6 pixel window, magnified

```
96 bits
```

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT	LF	VT	FF	CR	SO	SI
1	DLE	DC1	DC2	DC3	DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS	RS	US
2	SP	!	“	#	$	%	&	‘	(	)	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z	{	|	}	~	DEL

Hexadecimal to ASCII conversion table

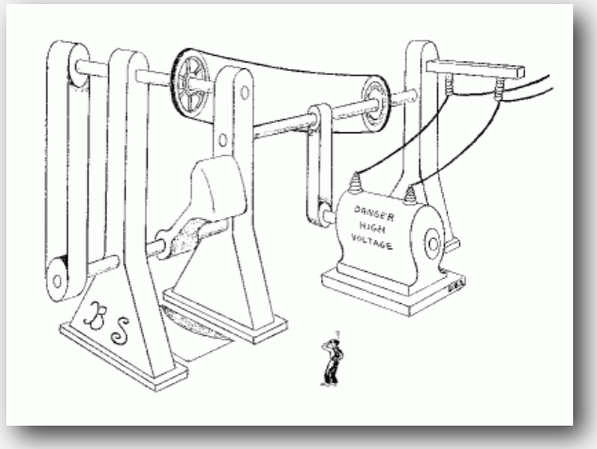
Universal data compression

US Patent 5,533,051 on "Methods for Data Compression", which is capable of compression all files.

Slashdot reports of the Zero Space Tuner™ and BinaryAccelerator™.

> *" ZeoSync has announced a breakthrough in data compression that allows for 100:1 lossless compression of random data. If this is true, our bandwidth problems just got a lot smaller.… "*

Physical analog. Perpetual motion machines.



Gravity engine by Bob Schadewald

Universal data compression

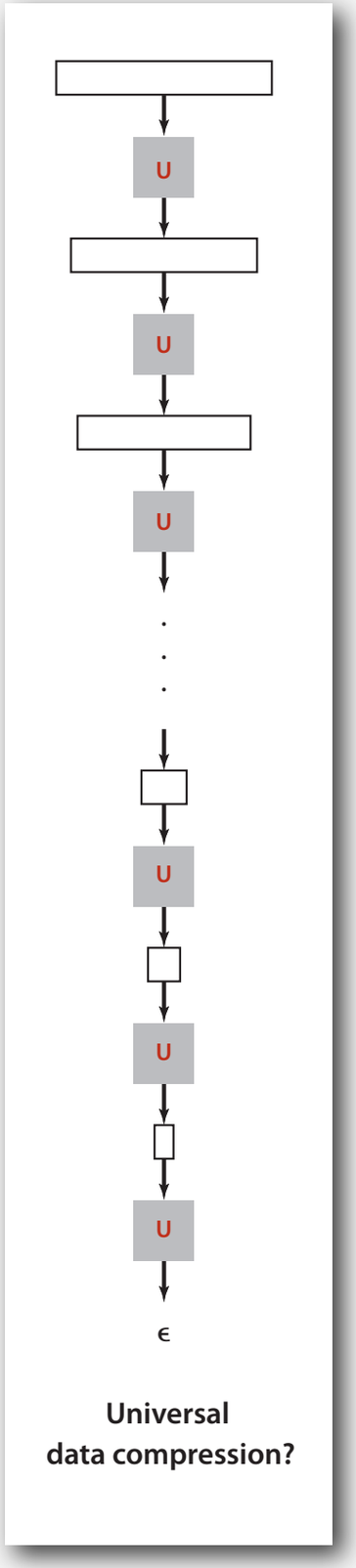
Proposition. No algorithm can compress every bitstring.

Pf 1. [by contradiction]

- Suppose you have a universal data compression algorithm U that can compress every bitstream.
- Given bitstring B_0, compress it to get smaller bitstring B_1.
- Compress B_1 to get a smaller bitstring B_2.
- Continue until reaching bitstring of size 0.
- Implication: all bitstrings can be compressed to 0 bits!

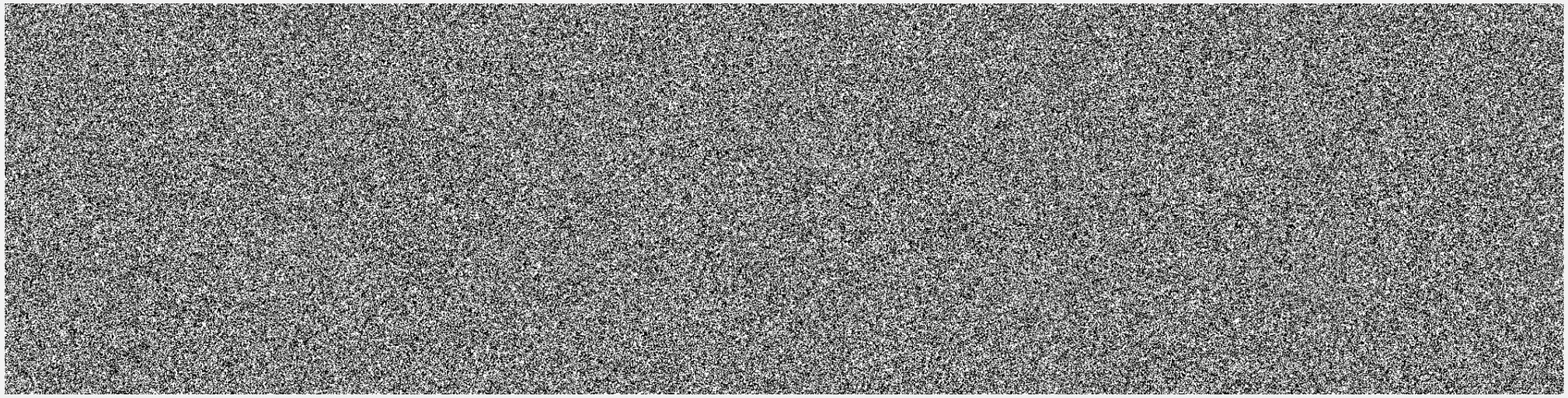
Pf 2. [by counting]

- Suppose your algorithm that can compress all $1{,}000$-bit strings.
- 2^{1000} possible bitstrings with $1{,}000$ bits.
- Only $1 + 2 + 4 + \ldots + 2^{998} + 2^{999}$ can be encoded with ≤ 999 bits.
- Similarly, only 1 in 2^{499} bitstrings can be encoded with ≤ 500 bits!



Undecidability

```
% java RandomBits | java PictureDump 2000 500
```



1000000 bits

A difficult file to compress: one million (pseudo-) random bits

```
public class RandomBits
{
   public static void main(String[] args)
   {
      int x = 11111;
      for (int i = 0; i < 1000000; i++)
      {
         x = x * 314159 + 218281;
         BinaryStdOut.write(x > 0);
      }
      BinaryStdOut.close();
   }
}
```

Rdenudcany in Enlgsih Inagugae

Q. How much redundancy is in the English language?

" ... randomising letters in the middle of words [has] little or no effect on the ability of skilled readers to understand the text. This is easy to denmtrasote. In a pubiltacion of New Scnieitst you could ramdinose all the letetrs, keipeng the first two and last two the same, and reibadailty would hadrly be aftcfeed. My ansaylis did not come to much beucase the thoery at the time was for shape and senqeuce retigcionon. Saberi's work sugsegts we may have some pofrweul palrlael prsooscers at work. The resaon for this is suerly that idnetiyfing coentnt by paarllel prseocsing speeds up regnicoiton. We only need the first and last two letetrs to spot chganes in meniang. " — *Graham Rawlinson*

A. Quite a bit

Rdenudcany in Turkish Inagugae

Q. How much redundancy is in the Turkish language?

> “ Bir İgnliiz Üvnseritsinede ypalaın arşaıtramya gröe, kleimleirn hrfalreiinn hnagi sırdaa yzalıdkılraı ömneli dğeliimş. Öenlmi oaln brincii ve snonucnu hrfain yrenide omlsaımyş. Ardakai hfraliren srısaı krıaşk oslada ouknyuorumş. Çnükü kleimlrei hraf hrafdğeil bri btün oalark oykuorumuşz” —*Anonymous*

A. Quite a bit

Data Compression

- Run-length coding
- Huffman compression

Run-length encoding

Simple type of redundancy in a bitstream. Long runs of repeated bits.

```
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1
```

40 bits

Representation. Use 4-bit counts to represent alternating runs of 0s and 1s: 15 0s, then 7 1s, then 7 0s, then 11 1s.

```
1 1 1 1 0 1 1 1 0 1 1 1 1 0 1 1
```

16 bits (instead of 40)

15 7 7 11

Q. How many bits to store the counts?

A. We'll use 8 (but 4 in the example above).

Q. What to do when run length exceeds max count?

A. If longer than 255, intersperse runs of length 0.

Applications. JPEG, ITU-T T4 Group 3 Fax, ...

Run-length encoding: Java implementation

```
public class RunLength
{
   private final static int R   = 256;
   private final static int lgR = 8;

   public static void compress()
   {  /* see textbook */  }

   public static void expand()
   {
      boolean bit = false;
      while (!BinaryStdIn.isEmpty())
      {
         int run = BinaryStdIn.readInt(lgR);
         for (int i = 0; i < run; i++)
            BinaryStdOut.write(bit);
         bit = !bit;
      }
      BinaryStdOut.close();
   }

}
```

maximum run-length count

number of bits per count

read 8-bit count from standard input

write 1 bit to standard output

pad 0s for byte alignment

An application: compress a bitmap

Typical black-and-white-scanned image.

- 300 pixels/inch.
- 8.5-by-11 inches.
- 300 × 8.5 × 300 × 11 = 8.415 million bits.

Observation. Bits are mostly white.

Typical amount of text on a page.

40 lines × 75 chars per line = 3,000 chars.

```
% java BinaryDump 32 < q32x48.bin

00000000000000000000000000000000    32
00000000000000000000000000000000    32
00000000000000011111110000000000    15  7 10
00000000000011111111111111100000    12 15  5
00000000001111000011111111100000    10  4  4  9  5
00000000111100000000011111100000     8  4  9  6  5
00000001110000000000001111100000     7  3 12  5  5
00000011110000000000001111100000     6  4 12  5  5
00000111100000000000001111100000     5  4 13  5  5
00001111000000000000001111100000     4  4 14  5  5
00001111000000000000001111100000     4  4 14  5  5
00011110000000000000001111100000     3  4 15  5  5
00011110000000000000001111100000     2  5 15  5  5
00111110000000000000001111100000     2  5 15  5  5
00111110000000000000001111100000     2  5 15  5  5
00111110000000000000001111100000     2  5 15  5  5
00111110000000000000001111100000     2  5 15  5  5
00111110000000000000001111100000     2  5 15  5  5
00111110000000000000001111100000     2  5 15  5  5
00111110000000000000001111100000     2  5 15  5  5
00111110000000000000001111100000     2  5 15  5  5
00111110000000000000001111100000     2  5 15  5  5
00111110000000000000001111100000     2  6 14  5  5
00111110000000000000001111100000     2  6 14  5  5
00011111100000000000001111100000     3  6 13  5  5
00011111100000000000001111100000     3  6 13  5  5
00001111110000000000001111100000     4  6 12  5  5
00000111111100000000001111100000     4  7 11  5  5
00000011111110000000001111100000     5  7 10  5  5
00000001111111100000000111111100000  6  8  7  6  5
00000000111111111111111111111100000  7 20  5
00000000001111111111001111111100000  9 11  2  5  5
00000000000001111100000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
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00000000000000000000000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
00000000000000000000000111111100000 22  5  5
00000000000000000000011111111110000 21  7  4
00000000000000000001111111111111100 18 12  2
00000000000000000011111111111111110 17 14  1
00000000000000000000000000000000000 32
00000000000000000000000000000000000 32

1536 bits
```

7 1s

17 0s

A typical bitmap, with run lengths for each row

Black and white bitmap compression: another approach

Fax machine (~1980).

- Slow scanner produces lines in sequential order.
- Compress to save time (reduce number of bits to send).

Electronic documents (~2000).

- High-resolution scanners produce huge files.
- Compress to save space (reduce number of bits to save).

Idea.

- use OCR to get back to ASCII (!)
- use Huffman on ASCII string (!)

Bottom line. Any extra information about file can yield dramatic gains.

Data Compression

- Run-length coding
- Huffman compression

Variable-length codes

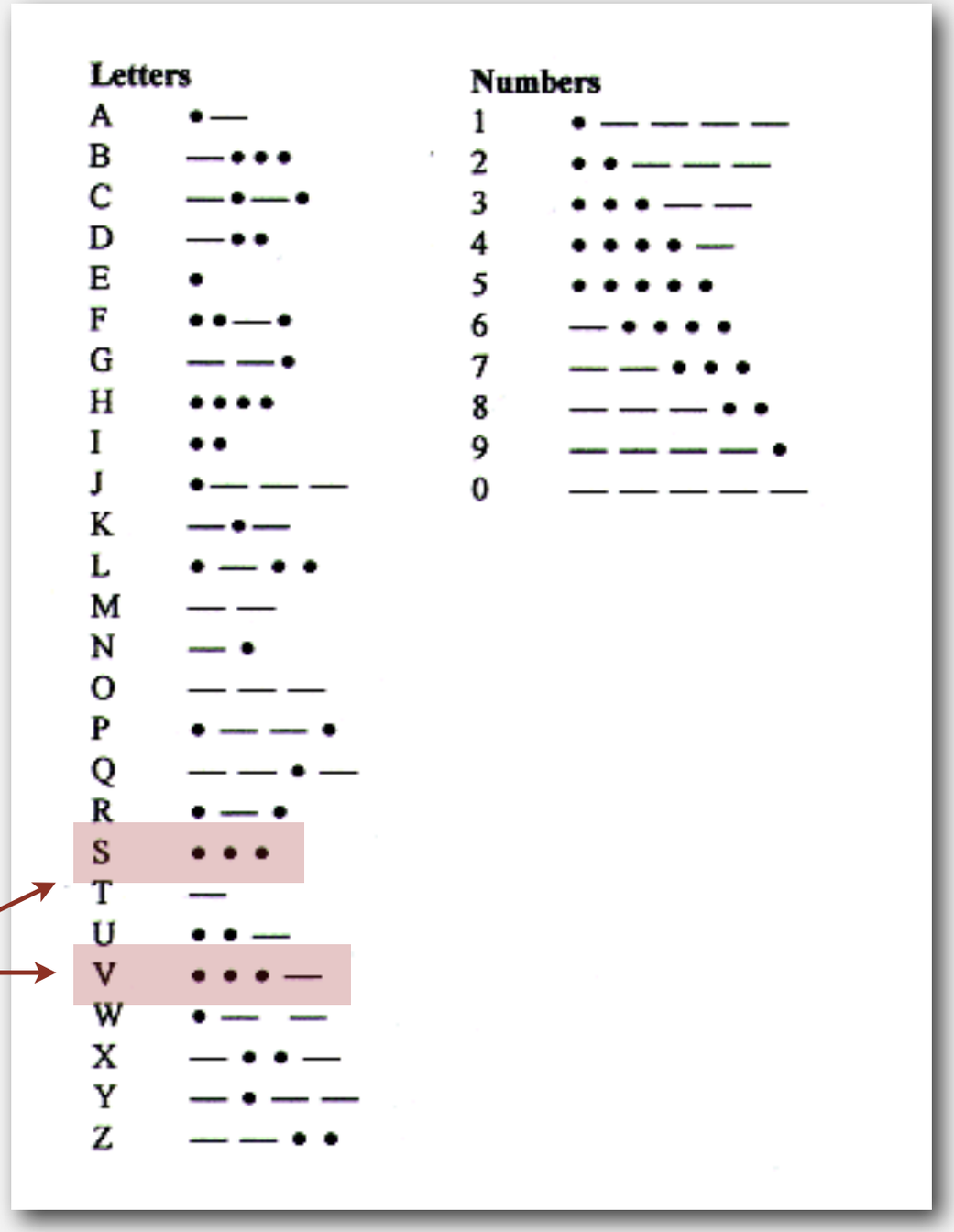
Use different number of bits to encode different chars.

Ex. Morse code: • • • – – – • • •

Issue. Ambiguity.

```
SOS ?
V7 ?
IAMIE ?
EEWNI ?
```

In practice. Use a medium gap to separate codewords.



Letters		Numbers	
A	•—	1	•—————
B	—•••	2	••———
C	—•—•	3	•••——
D	—••	4	••••—
E	•	5	•••••
F	••—•	6	—••••
G	——•	7	——•••
H	••••	8	———••
I	••	9	————•
J	•———	0	—————
K	—•—		
L	•—••		
M	——		
N	—•		
O	———		
P	•——•		
Q	——•—		
R	•—•		
S	•••		
T	—		
U	••—		
V	•••—		
W	•——		
X	—••—		
Y	—•——		
Z	——••		

codeword for S is a prefix of codeword for V

Variable-length codes

Q. How do we avoid ambiguity?

A. Ensure that no codeword is a prefix of another.

Ex 1. Fixed-length code.

Ex 2. Append special stop char to each codeword.

Ex 3. General prefix-free code.

Codeword table

key	*value*
!	101
A	0
B	1111
C	110
D	100
R	1110

Compressed bitstring

0111111100110010001111111100101 ← *30 bits*

A B RA CA DA B RA !

Codeword table

key	*value*
!	101
A	11
B	00
C	010
D	100
R	011

Compressed bitstring

11000111101011100110001111101 ← *29 bits*

A B R A C A D A B R A !

Prefix-free codes: trie representation

Q. How to represent the prefix-free code?

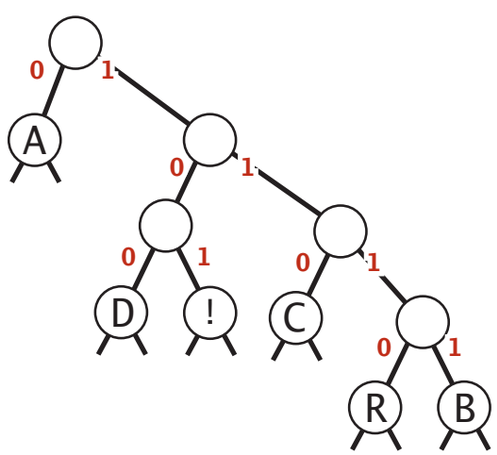
A. A binary trie!

- Chars in leaves.
- Codeword is path from root to leaf.

Codeword table

key	*value*
!	101
A	0
B	1111
C	110
D	100
R	1110

Trie representation



Compressed bitstring

01111111001100100011111100101 ← *30 bits*

A B RA CA DA B RA !

Codeword table

key	*value*
!	101
A	11
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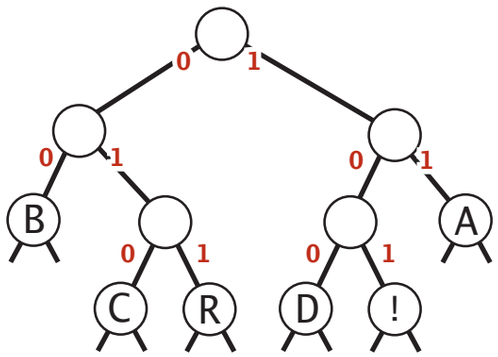
R	011

Trie representation



Compressed bitstring

1100011110101110011000111101 ← *29 bits*

A B R A C A D A B R A !

Prefix-free codes: compression and expansion

Compression.

- Method 1: start at leaf; follow path up to the root; print bits in reverse.
- Method 2: create ST of key-value pairs.

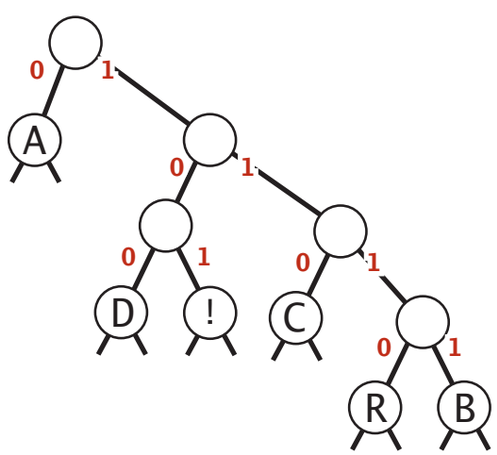
Expansion.

- Start at root.
- Go left if bit is 0; go right if 1.
- If leaf node, print char and return to root.

Codeword table

key	*value*
!	101
A	0
B	1111
C	110
D	100
R	1110

Trie representation



Compressed bitstring

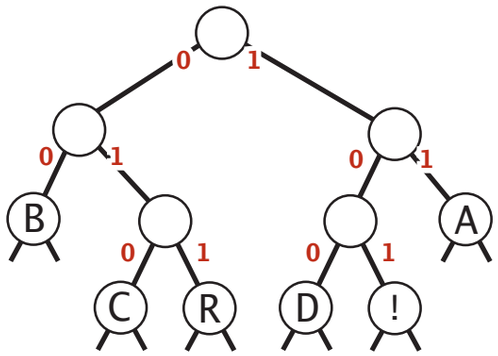
01111111001100100011111100101 ← *30 bits*

A B RA CA DA B RA !

Codeword table

key	*value*
!	101
A	11
B	00
C	010
D	100
R	011

Trie representation



Compressed bitstring

11000111101011100110001111101 ← *29 bits*

A B R A C A D A B R A !

Huffman trie node data type

```
private static class Node implements Comparable<Node>
{
   private char ch;    // Unused for internal nodes.
   private int freq;   // Unused for expand.
   private final Node left, right;

   public Node(char ch, int freq, Node left, Node right)
   {
      this.ch    = ch;
      this.freq  = freq;
      this.left  = left;
      this.right = right;
   }

   public boolean isLeaf()
   {  return left == null && right == null; }

   public int compareTo(Node that)
   {  return this.freq - that.freq;  }
}
```

initializing constructor

is Node a leaf?

compare Nodes by frequency (stay tuned)

Prefix-free codes: expansion

```
public void expand()
{
   Node root = readTrie();
   int N = BinaryStdIn.readInt();

   for (int i = 0; i < N; i++)
   {
      Node x = root;
      while (!x.isLeaf())
      {
         if (!BinaryStdIn.readBoolean())
            x = x.left;
         else
            x = x.right;
      }
      BinaryStdOut.write(x.ch, 8);
   }
   BinaryStdOut.close();
}
```

read in encoding trie

read in number of chars

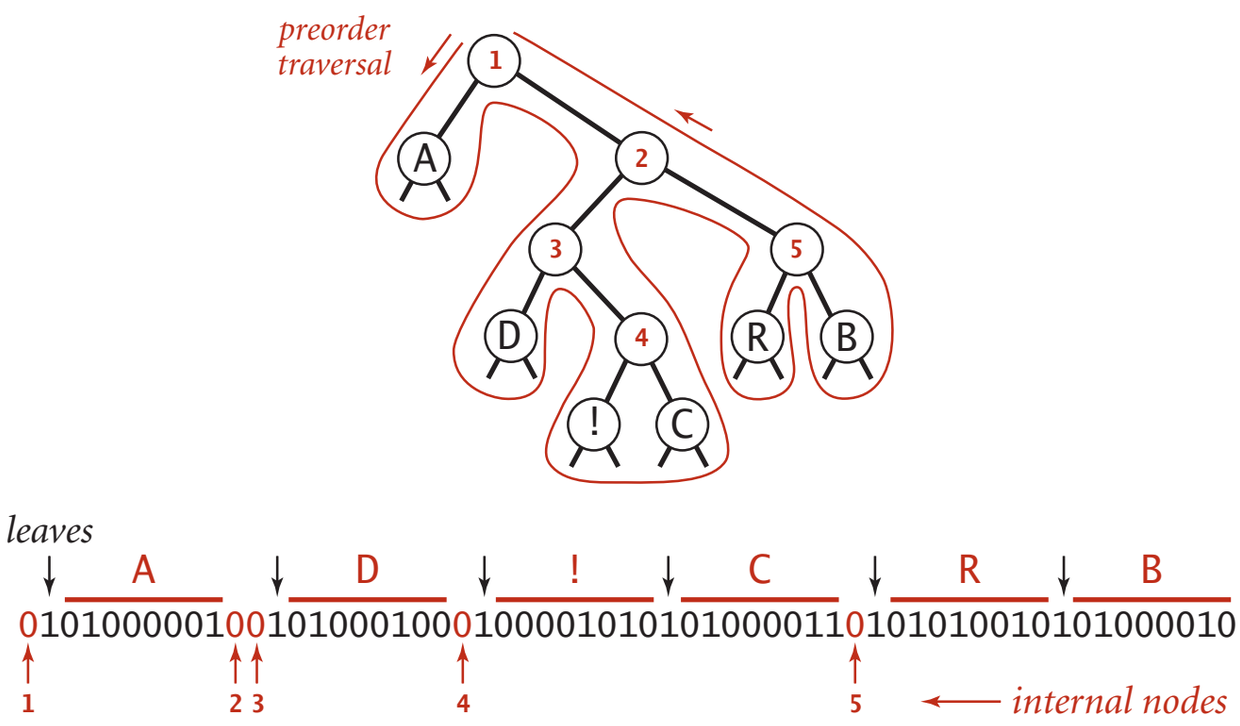
expand codeword for i^{th} char

Running time. Linear in input size N.

Prefix-free codes: how to transmit

Q. How to write the trie?

A. Write preorder traversal of trie; mark leaf and internal nodes with a bit.



Using preorder traversal to encode a trie as a bitstream

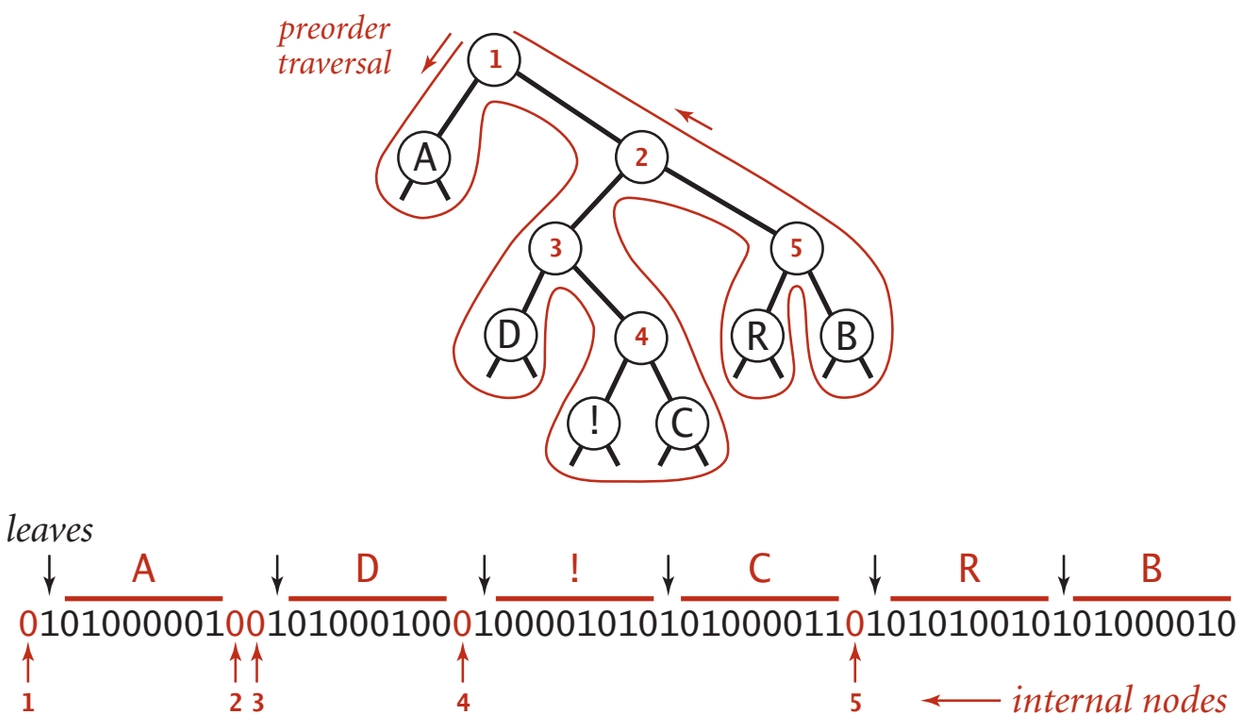
```
private static void writeTrie(Node x)
{
   if (x.isLeaf())
   {
      BinaryStdOut.write(true);
      BinaryStdOut.write(x.ch, 8);
      return;
   }
   BinaryStdOut.write(false);
   writeTrie(x.left);
   writeTrie(x.right);
}
```

Note. If message is long, overhead of transmitting trie is small.

Prefix-free codes: how to transmit

Q. How to read in the trie?

A. Reconstruct from preorder traversal of trie.



Using preorder traversal to encode a trie as a bitstream

```
private static Node readTrie()
{
   if (BinaryStdIn.readBoolean())
   {
      char c = BinaryStdIn.readChar(8);
      return new Node(c, 0, null, null);
   }
   Node x = readTrie();
   Node y = readTrie();
   return new Node('\0', 0, x, y);
}
```

not used for internal nodes

Shannon-Fano codes

Q. How to find best prefix-free code?

Shannon-Fano algorithm:

- Partition symbols S into two subsets S_0 and S_1 of (roughly) equal frequency.
- Codewords for symbols in S_0 start with 0; for symbols in S_1 start with 1.
- Recur in S_0 and S_1.

char	freq	encoding
A	5	0...
C	1	0...

S_0 = codewords starting with 0

char	freq	encoding
B	2	1...
D	1	1...
R	2	1...
!	1	1...

S_1 = codewords starting with 1

Problem 1. How to divide up symbols?

Problem 2. Not optimal!

Huffman algorithm

- Count frequency for each character in input.

char	freq	encoding
A	5	
B	2	
C	1	
D	1	
R	2	
!	1	

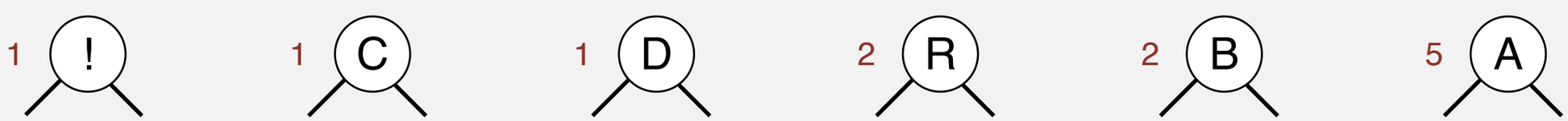
input

A B R A C A D A B R A !

Huffman algorithm

- Start with one node corresponding to each character with weight equal to frequency.

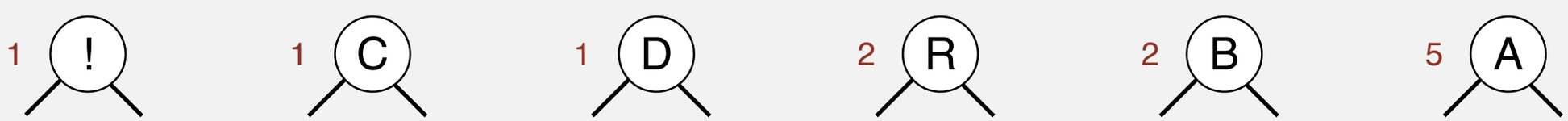
char	freq	encoding
A	5	
B	2	
C	1	
D	1	
R	2	
!	1	



Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

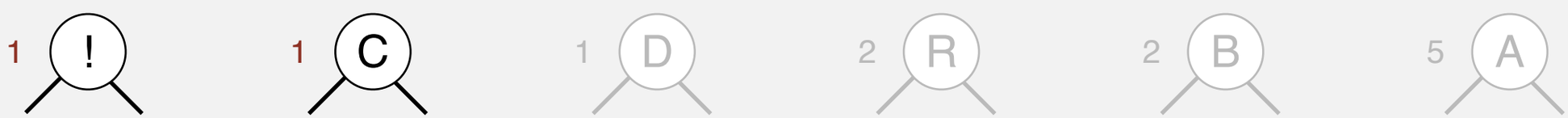
char	freq	encoding
A	5	
B	2	
C	1	
D	1	
R	2	
!	1	



Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

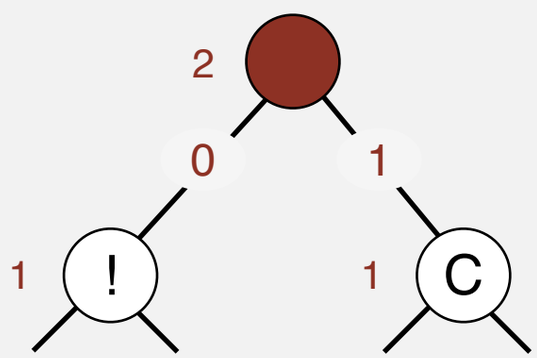
char	freq	encoding
A	5	
B	2	
C	1	
D	1	
R	2	
!	1	

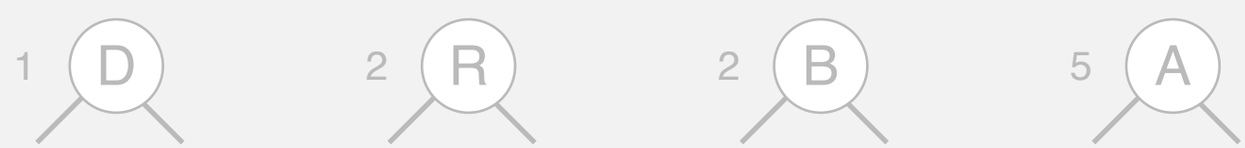


Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

char	freq	encoding
A	5	
B	2	
C	1	1
D	1	
R	2	
!	1	0

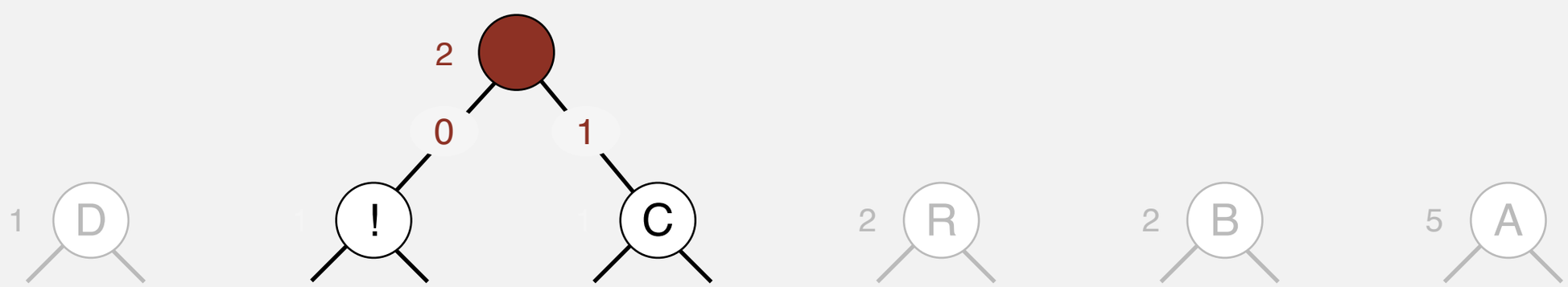




Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

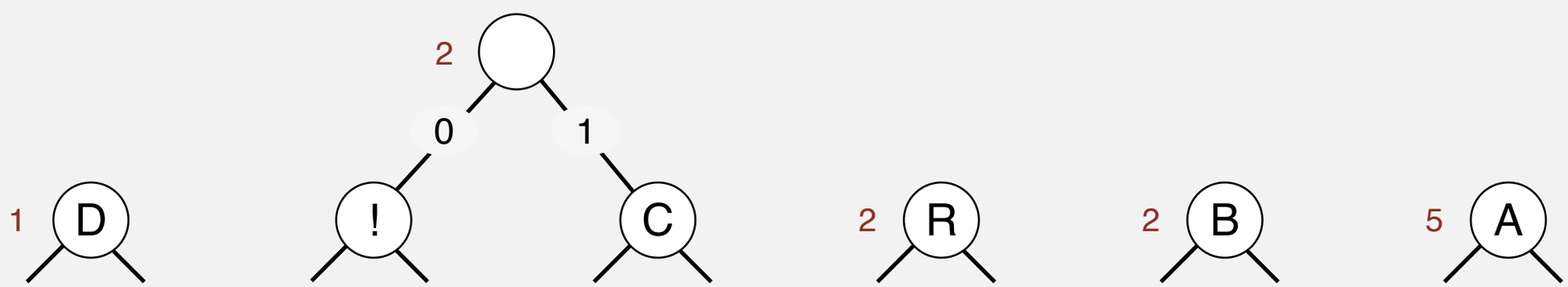
char	freq	encoding
A	5	
B	2	
C	1	1
D	1	
R	2	
!	1	0



Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

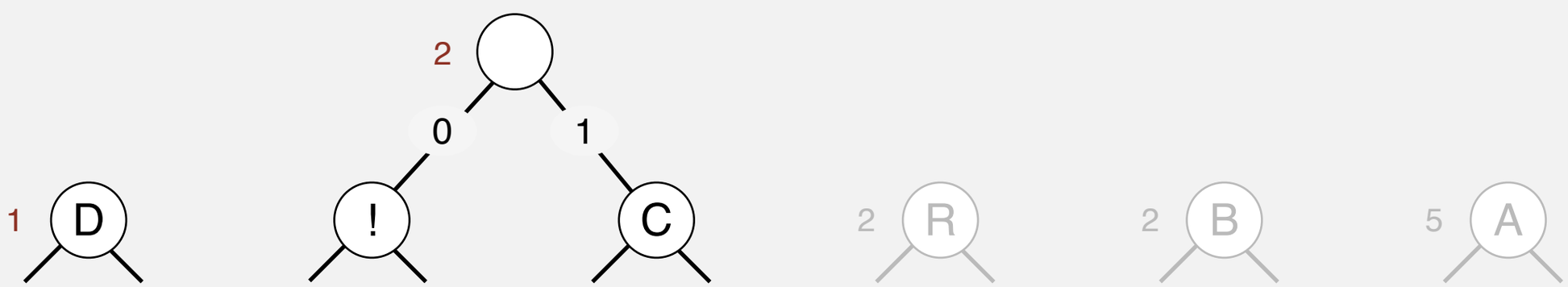
char	freq	encoding
A	5	
B	2	
C	1	1
D	1	
R	2	
!	1	0



Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

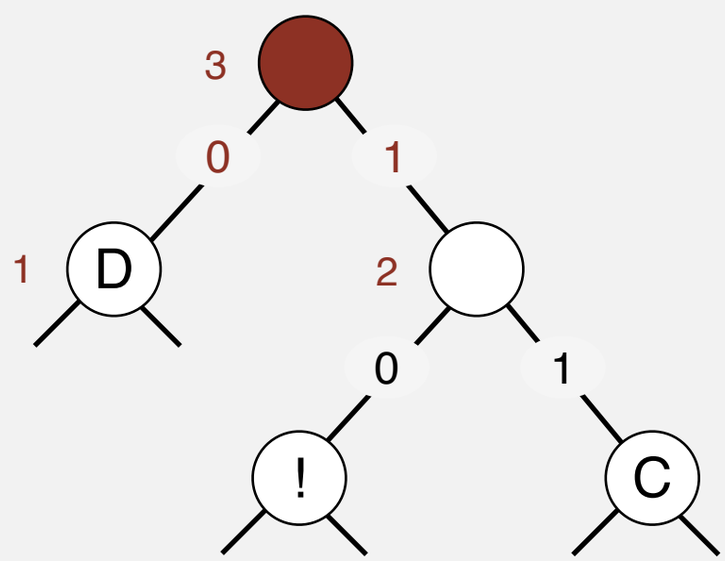
char	freq	encoding
A	5	
B	2	
C	1	1
D	1	
R	2	
!	1	0

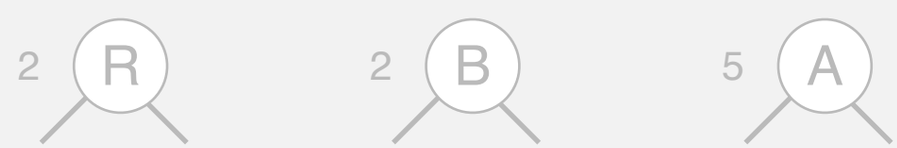


Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

char	freq	encoding
A	5	
B	2	
C	1	1 1
D	1	0
R	2	
!	1	1 0

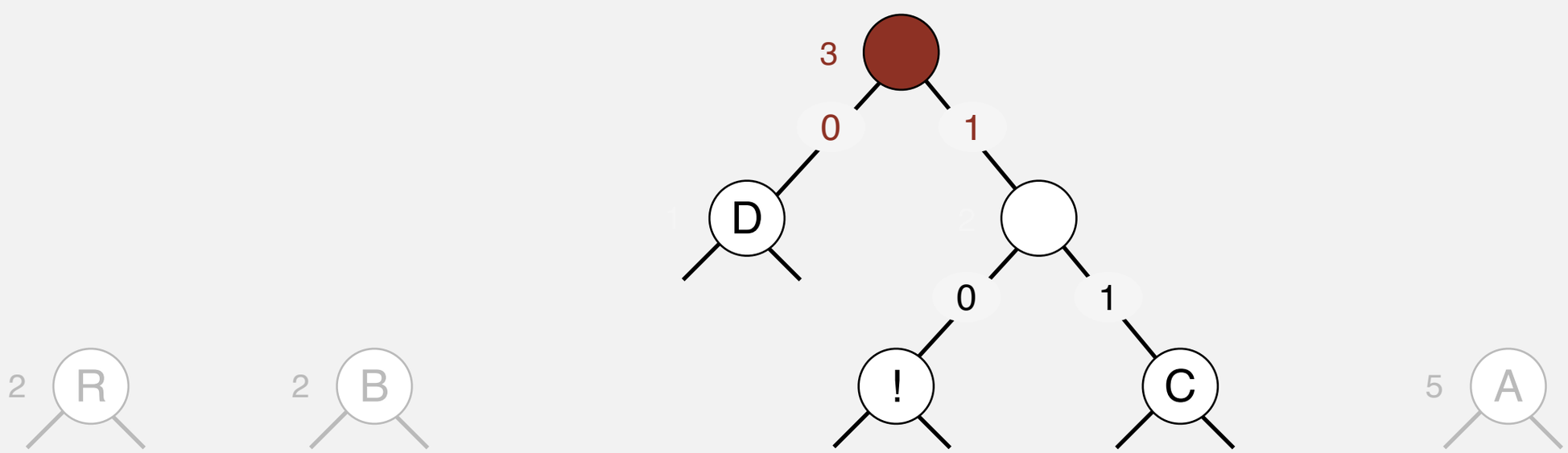




Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

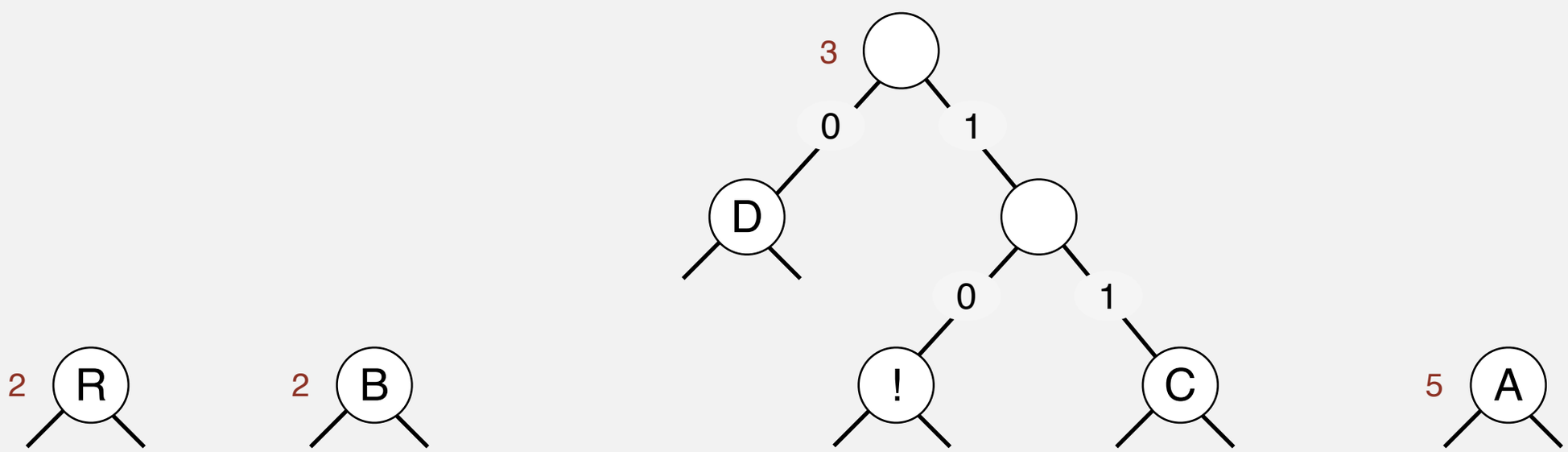
char	freq	encoding
A	5	
B	2	
C	1	1 1
D	1	0
R	2	
!	1	1 0



Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

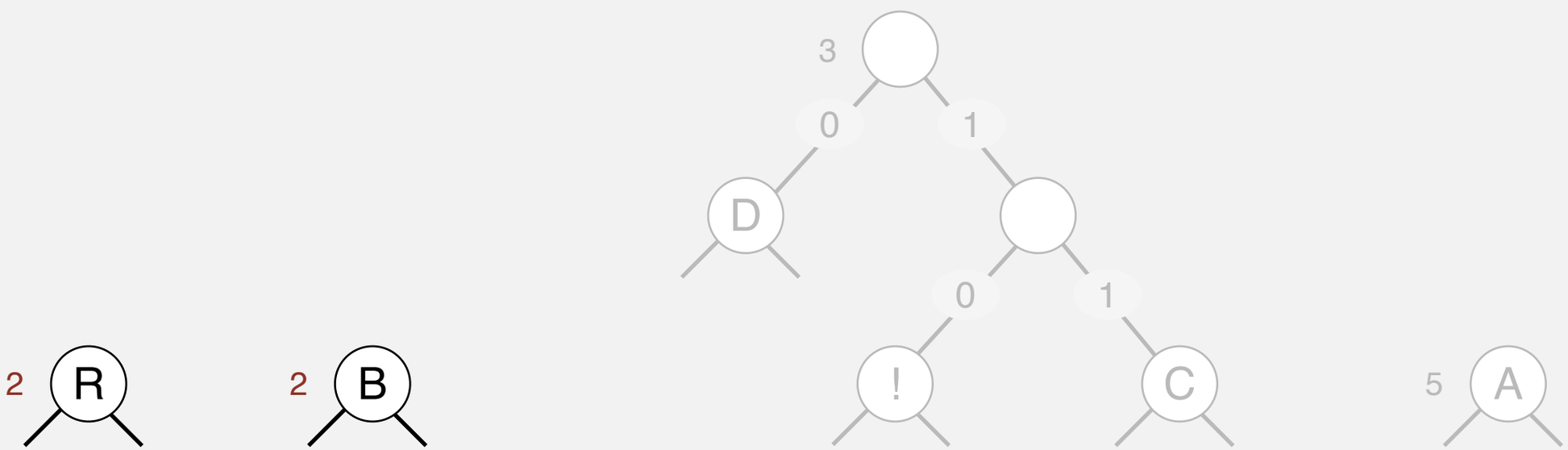
char	freq	encoding
A	5	
B	2	
C	1	1 1
D	1	0
R	2	
!	1	1 0



Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

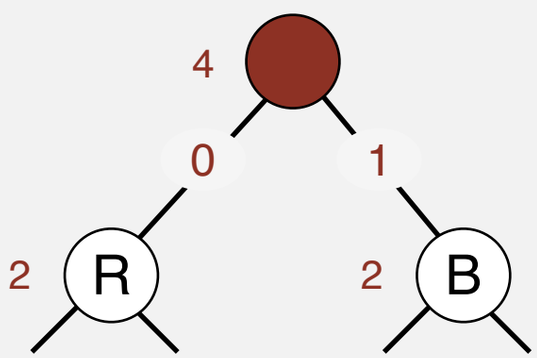
char	freq	encoding
A	5	
B	2	
C	1	1 1
D	1	0
R	2	
!	1	1 0

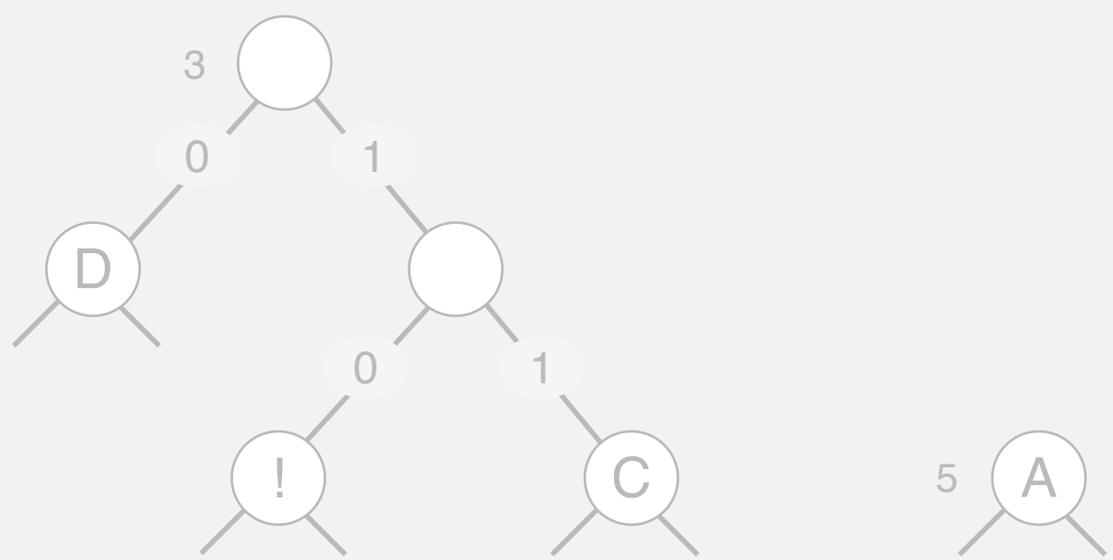


Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

char	freq	encoding
A	5	
B	2	1
C	1	1 1
D	1	0
R	2	0
!	1	1 0

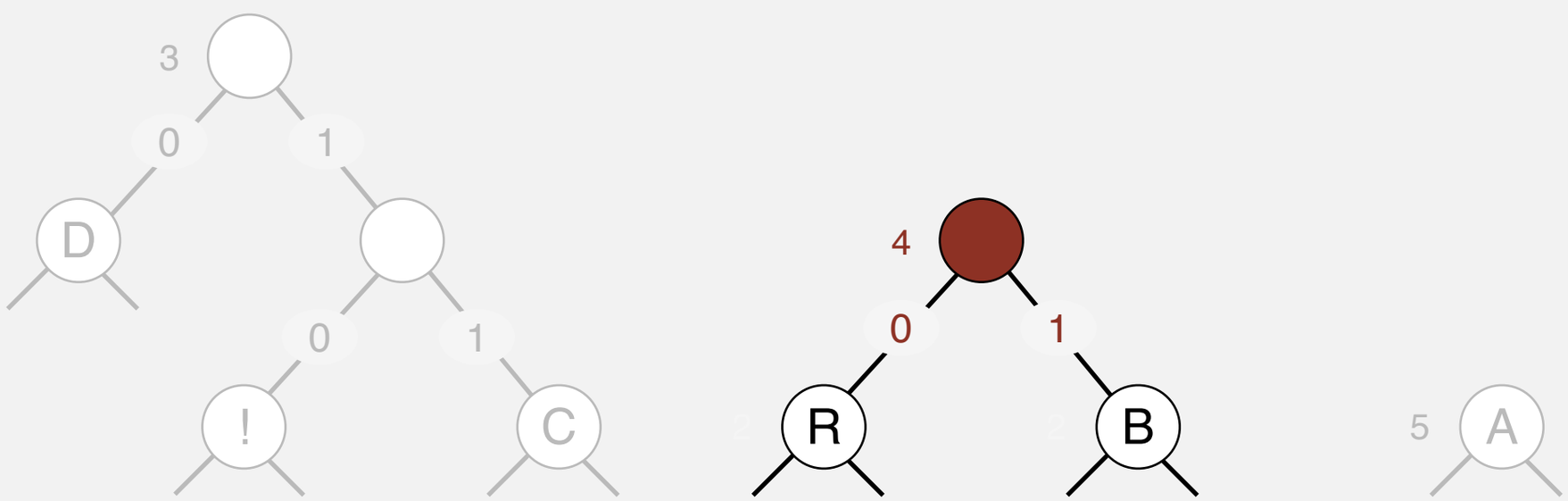




Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

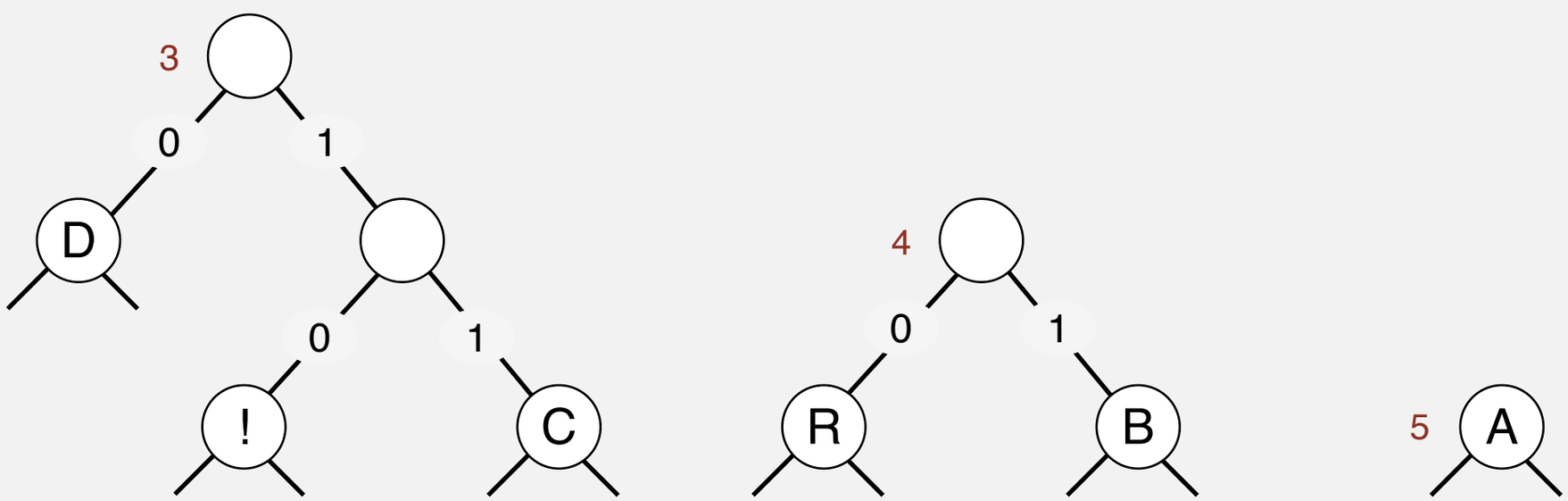
char	freq	encoding
A	5	
B	2	1
C	1	1 1
D	1	0
R	2	0
!	1	1 0



Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

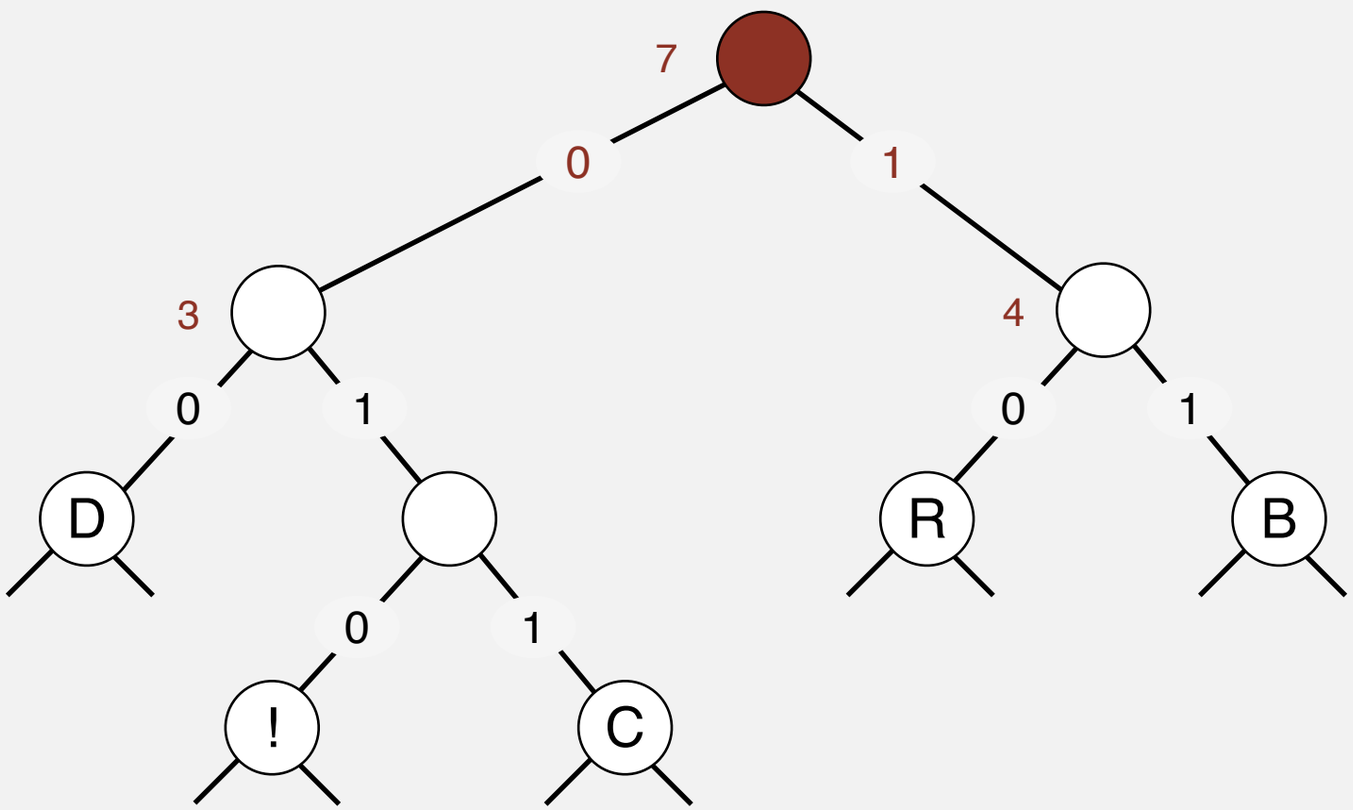
char	freq	encoding
A	5	
B	2	1
C	1	1 1
D	1	0
R	2	0
!	1	1 0



Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

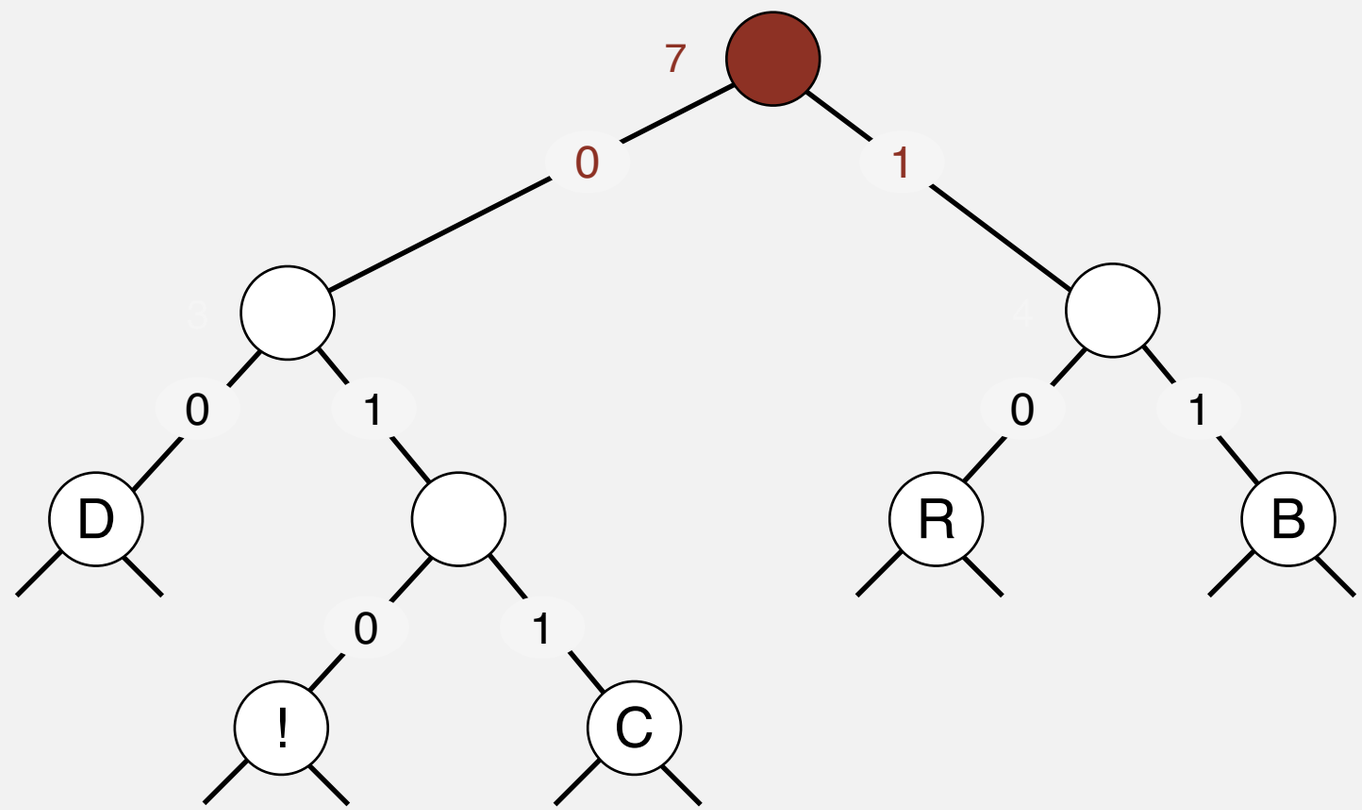
char	freq	encoding
A	5	
B	2	1 1
C	1	0 1 1
D	1	0 0
R	2	1 0
!	1	0 1 0



Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

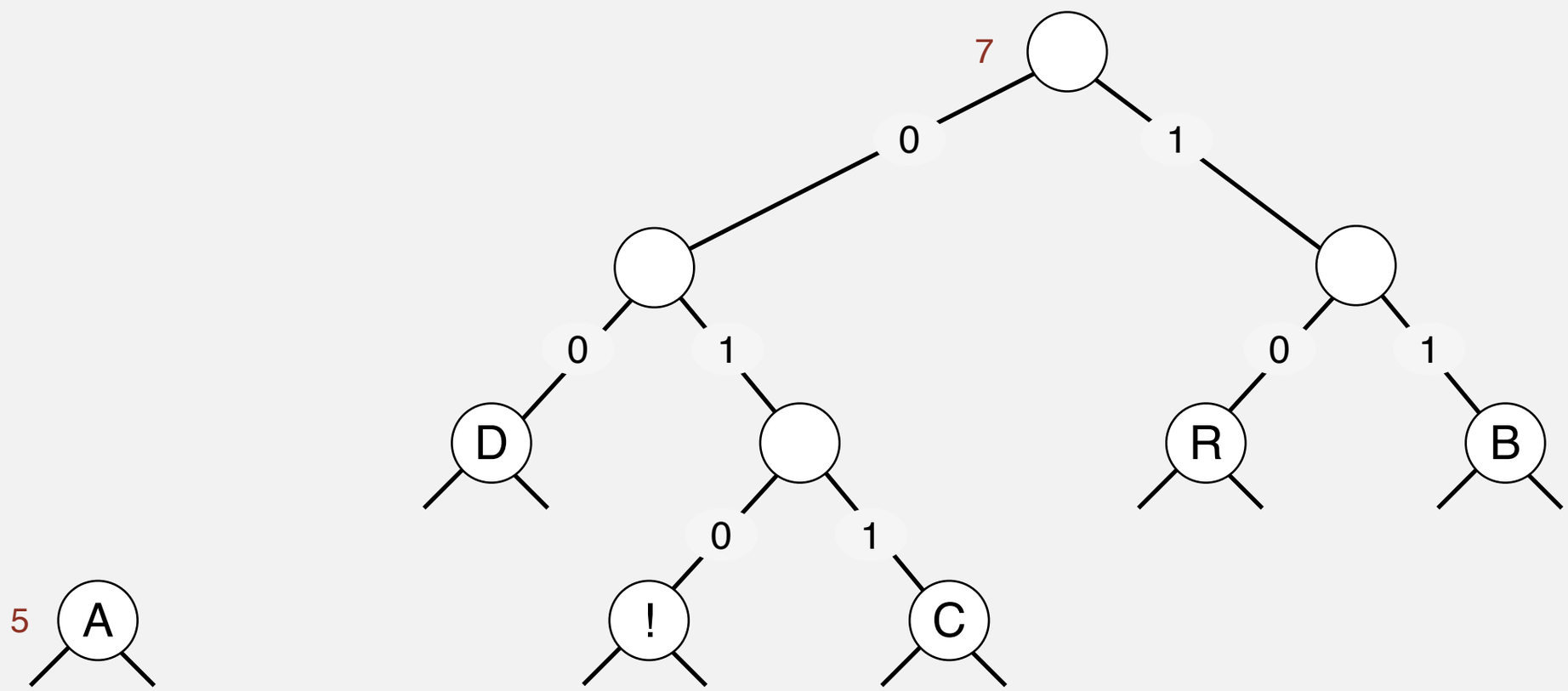
char	freq	encoding
A	5	
B	2	1 1
C	1	0 1 1
D	1	0 0
R	2	1 0
!	1	0 1 0



Huffman algorithm

- Select two tries with min weight.
- Merge into single trie with cumulative weight.

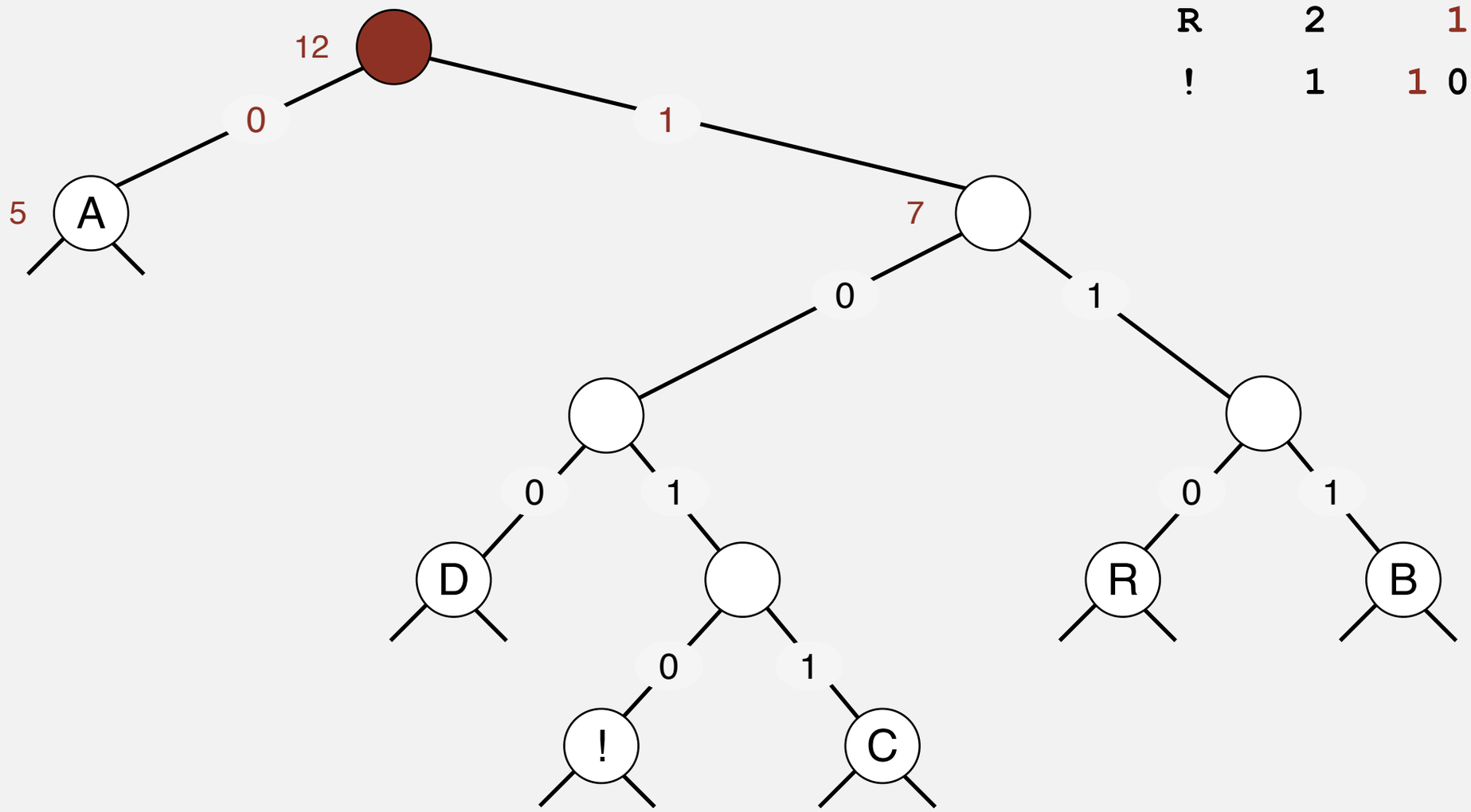
char	freq	encoding
A	5	
B	2	1 1
C	1	0 1 1
D	1	0 0
R	2	1 0
!	1	0 1 0



Huffman algorithm

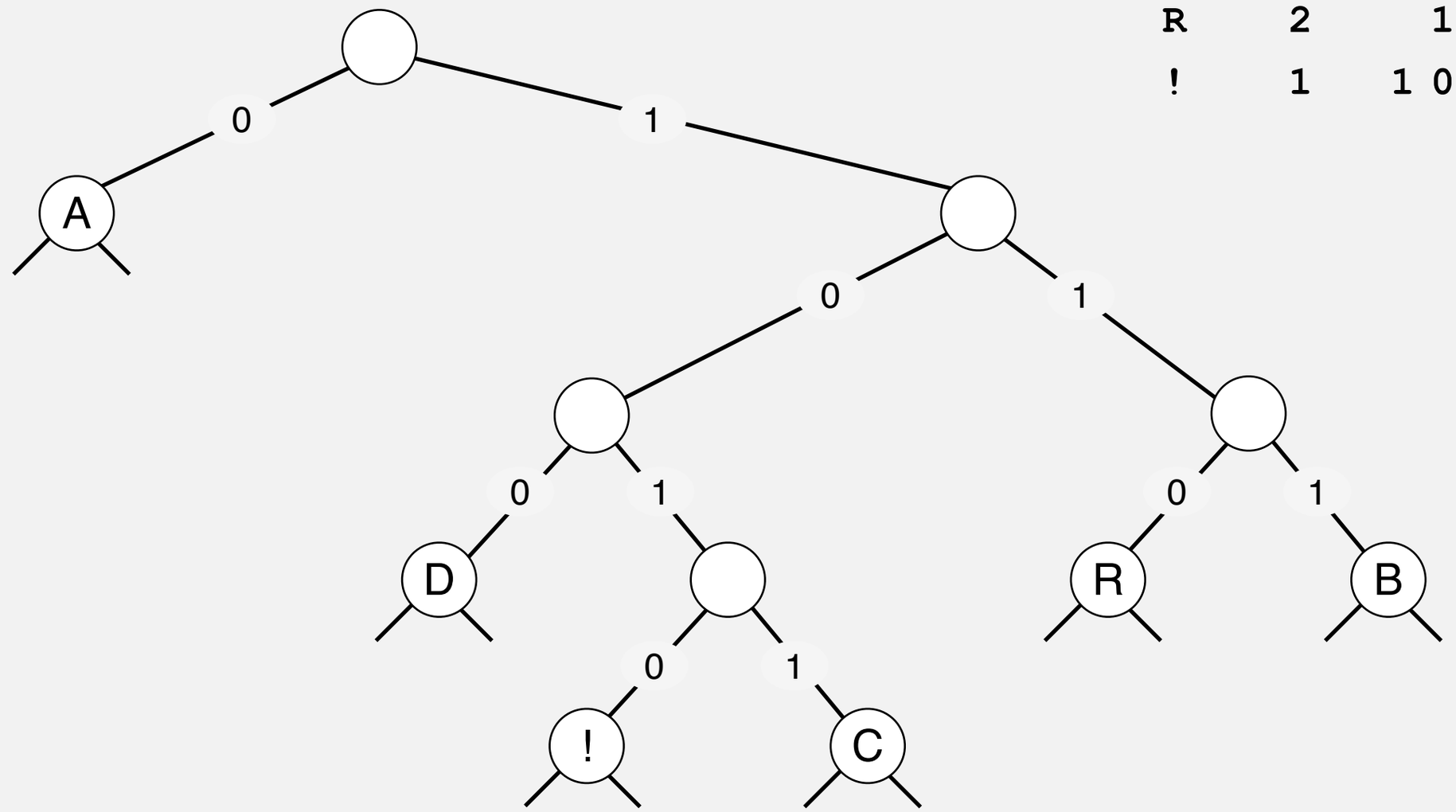
- Select two tries with min weight.
- Merge into single trie with cumulative weight.

char	freq	encoding
A	5	0
B	2	1 1 1
C	1	1 0 1 1
D	1	1 0 0
R	2	1 1 0
!	1	1 0 1 0



Huffman algorithm

char	freq	encoding
A	5	0
B	2	1 1 1
C	1	1 0 1 1
D	1	1 0 0
R	2	1 1 0
!	1	1 0 1 0



Huffman codes

Q. How to find best prefix-free code?

Huffman algorithm:

- Count frequency `freq[i]` for each char `i` in input.
- Start with one node corresponding to each char `i` (with weight `freq[i]`).
- Repeat until single trie formed:
  - select two tries with min weight `freq[i]` and `freq[j]`
  - merge into single trie with weight `freq[i] + freq[j]`

Applications:

Constructing a Huffman encoding trie: Java implementation

```
private static Node buildTrie(int[] freq)
{
    MinPQ<Node> pq = new MinPQ<Node>();
    for (char i = 0; i < R; i++)
       if (freq[i] > 0)
          pq.insert(new Node(i, freq[i], null, null));

    while (pq.size() > 1)
    {
       Node x = pq.delMin();
       Node y = pq.delMin();
       Node parent = new Node('\0', x.freq + y.freq, x, y);
       pq.insert(parent);
    }

    return pq.delMin();
}
```

initialize PQ with singleton tries

merge two smallest tries

not used for internal nodes

total frequency

two subtries

Huffman encoding summary

Proposition. [Huffman 1950s] Huffman algorithm produces an optimal prefix-free code.

no prefix-free code uses fewer bits

Pf. See textbook.

Implementation.

- Pass 1: tabulate char frequencies and build trie.
- Pass 2: encode file by traversing trie or lookup table.

Running time. Using a binary heap $\Rightarrow$ $N + R \log R$.

input size

alphabet size